CSL304 · ARTIFICIAL INTELLIGENCE · IIT BHILAI

Lab 2 — A* Search Assignment

Copying & Plagiarism Report

Students flagged	Basis
18 students, across 8 pairs/groups	Verbatim-quoted evidence from each student's own file

Prepared for instructor review.
Every quote below is copied exactly from the students' own notebooks — nothing is paraphrased.

1 How this report was built

We compared every student's write-up and code with every other student's, two at a time (4,753 pairs in total). This showed two very different kinds of matches. We had to tell them apart before flagging anyone.

> NOT FLAGGED — THIS KIND OF MATCH IS EXPECTED
>
> *"The assignment tells every student to reuse the same build_graph(), heuristic(), and astar_search() code, and the same Romania-map data. On top of that, about 51 students also use very similar general explanation text (things like "A* Search is an informed search algorithm..."). This text shows up across 50+ students who have nothing else in common, and their code does not match. So this looks like it came from the shared tutorial file, not from copying each other."*

> FLAGGED — THIS KIND OF MATCH IS NOT EXPECTED
>
> *"Some matches were found in only two (or a few) specific students, in parts that each student should have written themselves — their own personal notes about the notebook, their own choice of example scenario, or code formatting the assignment never asked for. These are the real red flags, and they are what the rest of this report is about."*

For every case below, we quote the exact matching text from each student's own file, so you can check it yourself. We also say clearly whether we could tell who copied from whom, or not.

2 Summary — 18 students flagged

Students	Concern	Evidence type
B24CS020 · B24ME007 · B24ME034 · B24CS045	High-confidence	Same personal write-up (4 students); one still has a wrong name in it
B24ME040 ↔ B24MT003	High-confidence	Same reworded intro + same code
B24EC013 ↔ B24EC023	High-confidence	Same personal write-up + same code
B24DS011 ↔ B24DS019	High-confidence	Same step-by-step notes + same code
B24CS039 ↔ B24DS030	High-confidence	Almost everything matches, even the typos — closest match found
B24CS006 ↔ B24MT001	High-confidence	Same personal write-up + same code
B24ME048 ↔ B24ME049	High-confidence	Same wording and typos in their own scenario write-up
B24DS036 ↔ B24DS506	Medium concern	Same code (down to formatting); write-ups look independent

3 Case-by-case evidence

CASE 1 HIGH-CONFIDENCE

Krrish Agarwal, Ayush Panwar, Patel Rudra Nitinkumar, Suyesh Singh

B24CS020 · B24ME007 · B24ME034 · B24CS045 — four-way group

All four students wrote the exact same "About this notebook" paragraph. This paragraph is written in the first person — it describes the student's own plan for the three tasks. A shared tutorial file would not contain something like this.

> THE SAME TEXT APPEARS IN ALL 4 FILES
> *"The helper functions build_graph(), heuristic(), draw_graph(), and astar_search() are kept exactly as given in the tutorial — no changes to their internal logic. What varies from task to task is only the data supplied to them: Task 1 uses the classic Romania map… Task 2 reuses the same map but deliberately inflates one heuristic value to test its effect. Task 3 works with a modified copy of astar_search(), where the priority-ordering key has been changed, per the assignment's explicit instructions."*

For one pair, we know exactly who copied from whom. B24ME034's own notebook still has this at the top:

> B24ME034'S TITLE CELL (THIS SHOULD SAY PATEL RUDRA NITINKUMAR / B24ME034)
> *"Krrish Agarwal B24CS020"*

This is clear proof: Patel Rudra Nitinkumar (B24ME034) copied Krrish Agarwal's (B24CS020) notebook, and forgot to change the name at the top. B24ME007 and B24CS045 do have their own correct names in their files, so we cannot say for sure who copied whom among CS020, ME007, and CS045 — only that the three of them share text that should have been written independently.

Markdown similarity: 0.87 – 1.00 across the group **Code similarity:** 1.000 across the group

CASE 2

Sanjay Swami ↔ Ayush Meena

HIGH-CONFIDENCE

B24ME040 ↔ B24MT003

Both students wrote the standard A* introduction in their own words, instead of copying it from the tutorial. But their two "own words" versions are identical to each other — and different from the wording the other 51-student group uses. So this cannot be explained by "everyone got it from the tutorial."

> THE SAME TEXT APPEARS IN BOTH FILES
> *"A* chooses nodes using both the cost already travelled and an estimate of the remaining cost. … This is exactly like building an adjacency list from an edge array in C — two explicit loops, no shortcuts."*

We cannot tell who copied from whom here — nothing in either file points to one student writing it first.

Markdown similarity: 1.000 **Code similarity:** 1.000

CASE 3

Kilaparthi Parveen Satya Teja ↔ Peeriga Rakesh

HIGH-CONFIDENCE

B24EC013 ↔ B24EC023

THE SAME TEXT APPEARS IN BOTH FILES

"Same code and structure as the class tutorial (build_graph(), heuristic(), draw_graph(), astar_search()). I only changed the data (nodes/edges) to match the Romania map, and made small, specific tweaks where each task asked for it."

This is written in the first person — the student describing their own approach. It is word-for-word the same in two students' files who otherwise have nothing to do with each other.

Markdown similarity: 1.000 **Code similarity:** 1.000

CASE 4

Kollati Devi Chandrika ↔ Namburi Veda Aishwarya

B24DS011 ↔ B24DS019

HIGH-CONFIDENCE

THE SAME TEXT APPEARS IN BOTH FILES

"Step 1 — Romania map node coordinates: These are the coordinates supplied in the assignment and are used by draw_graph(). … Step 3 — Build the adjacency list: This is the same build_graph() structure as the tutorial."

We cannot tell who copied from whom here.

Markdown similarity: 1.000 **Code similarity:** 1.000

CASE 5

Shivam Chaurasia ↔ Siddhesh Bansal

B24CS039 ↔ B24DS030 — strongest match found in the whole scan

HIGH-CONFIDENCE

This is the closest match we found anywhere. The write-up, the code, and even the spelling mistakes all match — including the odd phrase "changed their name so that no problem is created", which appears word-for-word in both Task 2 and Task 3 of both notebooks, and the same misspelling of "source" as "sourse".

TASK 2, SAME WORDS IN BOTH FILES

"Note- I have just changed the function a little and just changed their name so that no problem is created"

TASK 2 EXPLANATION, SAME WORDS IN BOTH FILES

"So here we used f(n)=g(n)+h(n) where we replaced h(n) with Manhattan distance. Here the heuristic is inadmissible because it overestimates the minimum cost from node to goal, which violates h(n) <= h(n). We can clearly see this as the heuristic gives a minimum distance of 450 km but our optimal distance was 418 km."*

TASK 3 EXPLANATION, SAME WORDS IN BOTH FILES (SAME "SOURSE" TYPO)

"There will be no incomplete case as when we use h(n) here as heuristic function and goal as bucharest then whichever node we take there will be a neighbour with less h(n) and it never gets stuck in loop… But if we change destination or goal as Oradea and sourse as Iasi then we get stuck in a loop of Iasi and Neamt where we

> *never reach goal”*

We cannot tell who copied from whom — almost the whole notebook matches, so either student could be the original.

Markdown similarity: 1.000 **Code similarity:** 1.000 (all 3 required functions)

CASE 6

Anshul Singh ↔ Aman Saini

HIGH-CONFIDENCE

B24CS006 ↔ B24MT001

THE SAME TEXT APPEARS IN BOTH FILES

> *“This notebook is completed from the supplied A* Search tutorial notebook and follows the assignment requirement to retain the tutorial's overall structure and functions.”*

We cannot tell who copied from whom here.

Markdown similarity: 1.000 **Code similarity:** 1.000

CASE 7

Tarun Yadav ↔ Tript Singh

HIGH-CONFIDENCE

B24ME048 ↔ B24ME049

This match is different from the big 51-student group. Here, the match is in each student's own scenario write-up for Task 2 and Task 3. They picked different cities and different coordinate changes — but the reasoning and the wording around their choices is almost the same.

TASK 3, ALMOST THE SAME WORDS IN BOTH FILES

> *“removed g(n) = new_g from new_f thus making f(n)=h(n)”*

B24ME048

> *“The solution is complete as we are using visited array in the function causing greed algorithm to be complete but it is not optimal”*

B24ME049 (ALMOST THE SAME, WITH A COUPLE OF EXTRA TYPOS)

> *“The solution is complete as we are using visited array inn the function causing greedy algorithm to be complete but it i s not optimal”*

We cannot tell who copied from whom here.

Markdown similarity: 0.946 **Code similarity:** 0.99

CASE 8

MEDIUM CONCERN

Vatsal Yadav ↔ Rudra Mehul Dudhat

B24DS036 ↔ B24DS506

This case is different from the ones above. These two students clearly wrote their own explanations (only 0.76 similarity, different style and structure). But their code is exactly the same, letter for letter — including a formatting choice (writing the city coordinates 3 per line) that the assignment did not ask for, and that no other student uses. This looks like shared or copied code, but a written-up-on-their-own notebook — so we mark it Medium concern, not High-confidence.

> **SAME FORMATTING IN BOTH FILES**
> *"nodes = { 'Oradea': (131, 571), 'Zerind': (108, 531), 'Arad': (91, 492), 'Timisoara': (94, 410), 'Lugoj': (165, 379), 'Mehadia': (168, 339), ..."*

We cannot tell who copied from whom here.

Markdown similarity: 0.764 **Code similarity:** 1.000

4 What we chose not to flag, and why

- About 51 students share the same general A* explanation and step-by-step text with C-language comparisons. This shows up across too many students who have nothing else in common, and their code does not match — so it looks like it came from the shared tutorial file, not from each other.
- Almost all students have near-identical build_graph(), heuristic(), and astar_search() code. The assignment tells them to reuse this code, so this is expected.
- Almost all students have the same Romania map data (city coordinates, road distances, heuristic values). The assignment gives them this data, so this is expected.
- Almost all students get the same correct answer (418 km, Arad → Sibiu → Rimnicu Vilcea → Pitesti → Bucharest). This is just the one correct answer — it does not mean they copied each other.